Gene Expression Prediction of Cortical Motor Neuron Morphology

Georgia Goddard

Interview 2 for the Research Assistant in Bioinformatics role (King's College London)

30/10/2026

Gene Expression Prediction of Cortical Motor Neuron Morphology

Task:

1. Identify genes whose expression changes correspond with changes in cell morphology
2. Represent the potential function of these genes relative to cell morphology
3. Provide statistical support

Gene Expression Prediction of Cortical Motor Neuron Morphology

```
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├── AI_use_statement.docx
├── Task2_presentation.pdf
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│   ├── m1_patchseq_morph_zprofiles.csv    # Neuron z-profiles
│   ├── m1_patchseq_morph_features.csv     # Neuron morphological features (numerical)
│   ├── m1_patchseq_exon_counts.csv.gz     # Neuron exon count matrix
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Data

Initial Dataset

>2000 cells processed:
- **1329** passed initial recording QC
- **646** complete morphologies

↓

Round 1 + 2 Cell Typing

Nearest-centroid classifier with **Pearson correlation** of top variable genes, against reference datasets

Bootstrapping with replacement for confidence

↓

Final Dataset

1227 final cells with t-type assignment:
- **85%** t-type coverage
- **52%** morphology coverage

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Final Dataset

1227 final cells with t-type assignment:
- **85%** t-type coverage
- **52%** morphology coverage

- **Meta-data (n = 1329)**

  `m1_patchseq_meta_data.csv`

- **Exon + Intron Counts (n = 1329)**

  `m1_patchseq_exon_counts.csv.gz`

  `m1_patchseq_intron_counts.csv.gz`

- **Morphological Features (n = 646)**

  `m1_patchseq_morph_zprofiles.csv`

  `m1_patchseq_morph_features.csv`

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Interrogate Data & Sample Exclusion

Metadata:

- "Exclusion reasons" column: 7 samples
- GitHub README: 1 sample with incorrect morphological reconstruction (20171207sample1)

Scala et al. methodology:

- t-type assignment: excludes a further **95** cells
- **Limitation:** SWC files unavailable – cells unidentifiable

Initial Dataset

```
exon_counts: (42466, 1329)
intron_counts: (42466, 1329)
morph_features: (646, 63)
morph_zprofiles: (646, 20)
```


Final Dataset

```
exon_counts: (42466, 1322)
intron_counts: (42466, 1322)
morph_features: (645, 63)
morph_zprofiles: (645, 20)
```

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Morphological Features

Morphological Features Excluded:

- Z-features
- Normalized depth
- Normalized layer (non-numerical)
- Cell class (neuron label - non-numerical)

Inhibitory Neurons (n = 371)

34 morphological features

Excitatory Neurons (n= 274)

49 morphological features

```
morph_features.columns
```

```
Index(['"apical" branch points', '"apical" height',
       '"apical" log1p number of outer bifurcations',
       '"apical" mean bifurcation distance', '"apical" robust height',
       '"apical" robust width', '"apical" std bifurcation distance',
       '"apical" total length', '"apical" width', 'EMD axon dendrite',
       'Log1p fraction of axon above dendrite',
       'Log1p fraction of axon below dendrite',
       'Log1p fraction of dendrite above axon', 'axon above soma',
       'axon bifurcation standard deviation', 'axon branch points',
       'axon first bifurcation moment', 'axon height',
       'axon log min tortuosity', 'axon max Euclidean dist',
       'axon max branch order', 'axon max path distance to soma',
       'axon max segment length', 'axon mean neurite radius',
       'axon min branch angle', 'axon robust height', 'axon robust width',
       'axon tips', 'axon total length', 'axon width', 'axon x-bias',
       'axon z-bias', 'dendrite above soma',
       'dendrite bifurcation standard deviation', 'dendrite branch points',
       'dendrite first bifurcation moment', 'dendrite height',
       'dendrite log max tortuosity', 'dendrite log median tortuosity',
       'dendrite log min tortuosity', 'dendrite max Euclidean dist',
       'dendrite max branch order', 'dendrite max path distance to soma',
       'dendrite max segment length', 'dendrite mean neurite radius',
       'dendrite median path angle', 'dendrite min branch angle',
       'dendrite robust height', 'dendrite robust width', 'dendrite tips',
       'dendrite total length', 'dendrite width', 'dendrite x-bias',
       'dendrite z-bias', 'mean initial segment radius', 'normalized depth',
       'soma radius', 'stems', 'stems exiting down',
       'stems exiting to the sides', 'stems exiting up', 'normalized layer',
       'cell class'],
      dtype='str')
```

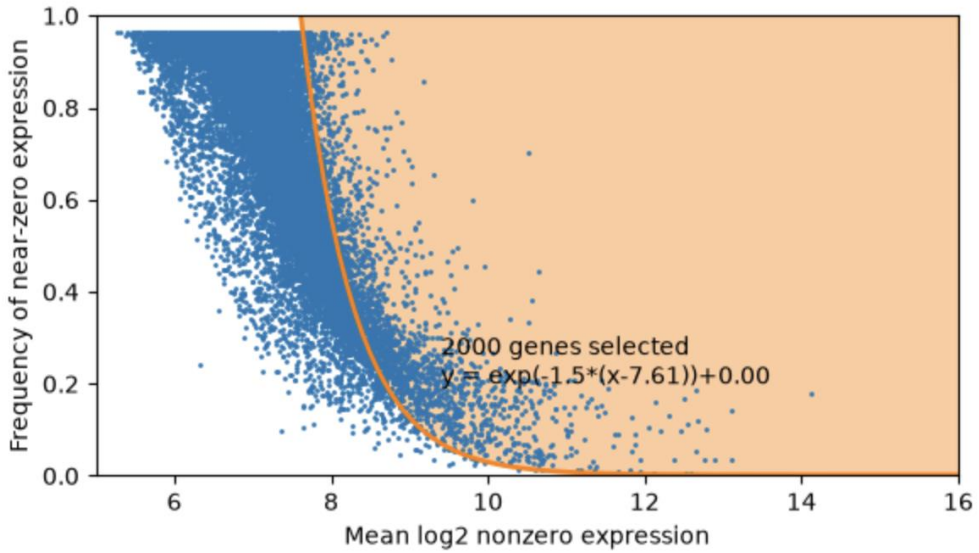
Identify Top Variable Genes

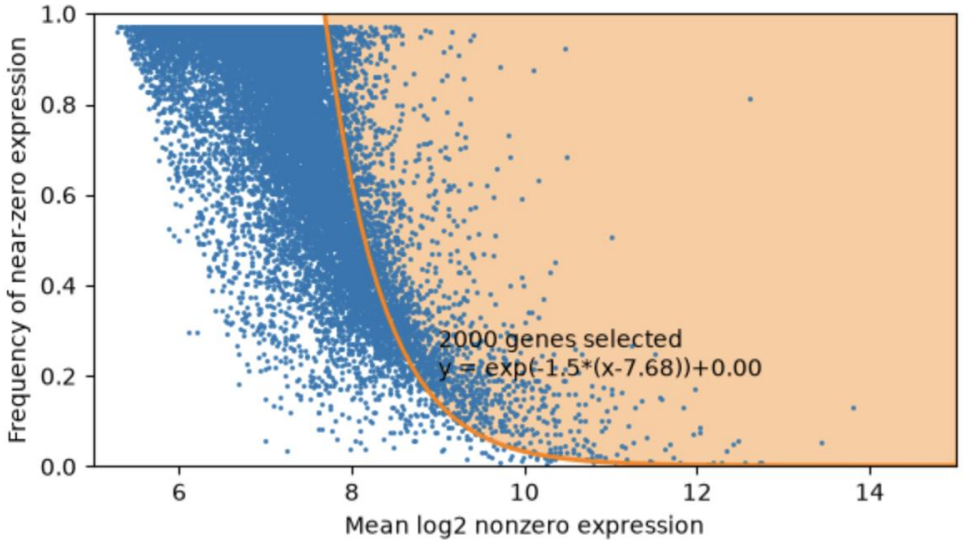
Methodology

- Split gene matrices by cell class
- Combined intron and exon counts
- Identify top 2000 variable genes in each dataset

rnaseqTools.py

```
exc_selected_genes = geneSelection(
    data=exc_matrix.values,
    threshold=32,
    atleast=10,
    yoffset=0.002,
    decay=1.5,
    n=2000,
    plot=True,
)
```





Expression and Morphology PCA

Methodology

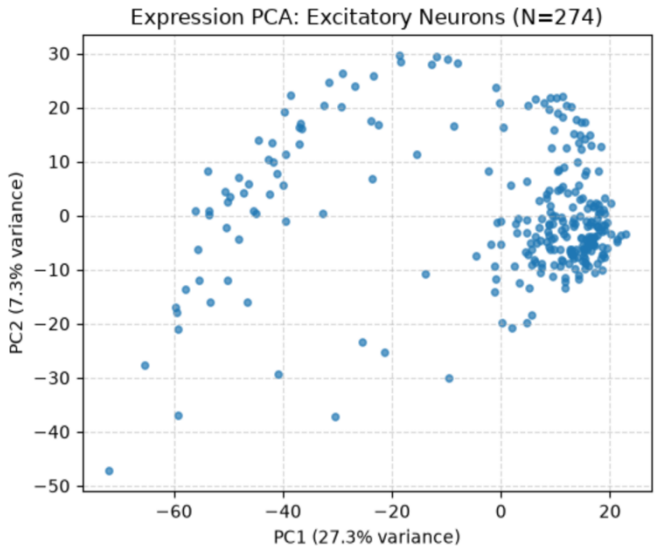
Top 2000 variable genes per cell class

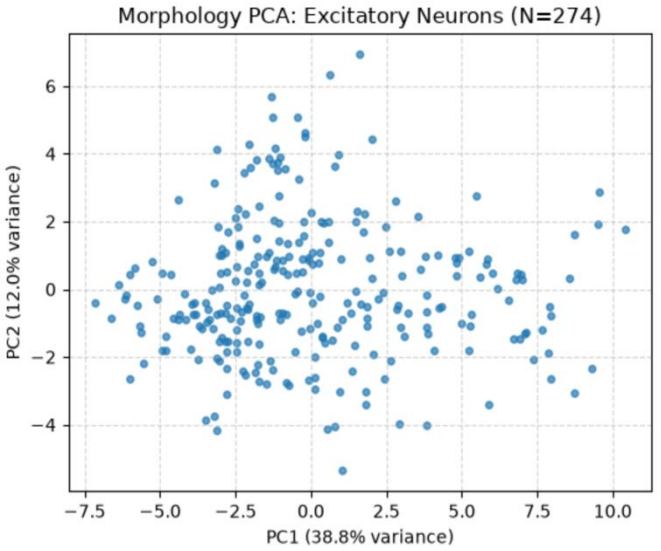
- Normalised & log2(x+1) transformed

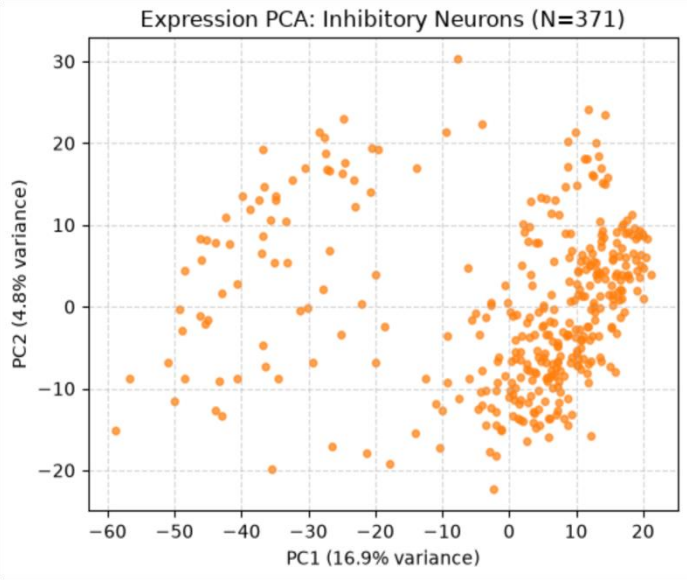
Morphological variables (exc = 34, inh = 49)

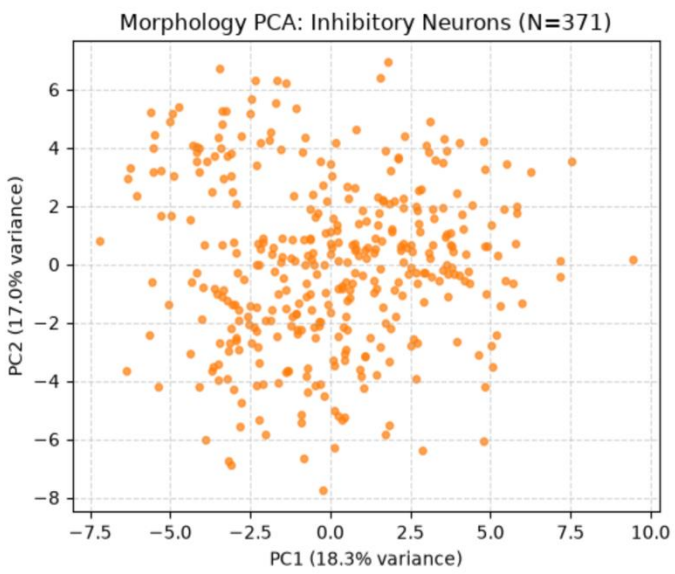
- Handle missing values

=> Z-score scaling









Pairwise PC Correlation

Methodology

- Permutation testing (n = 1000)

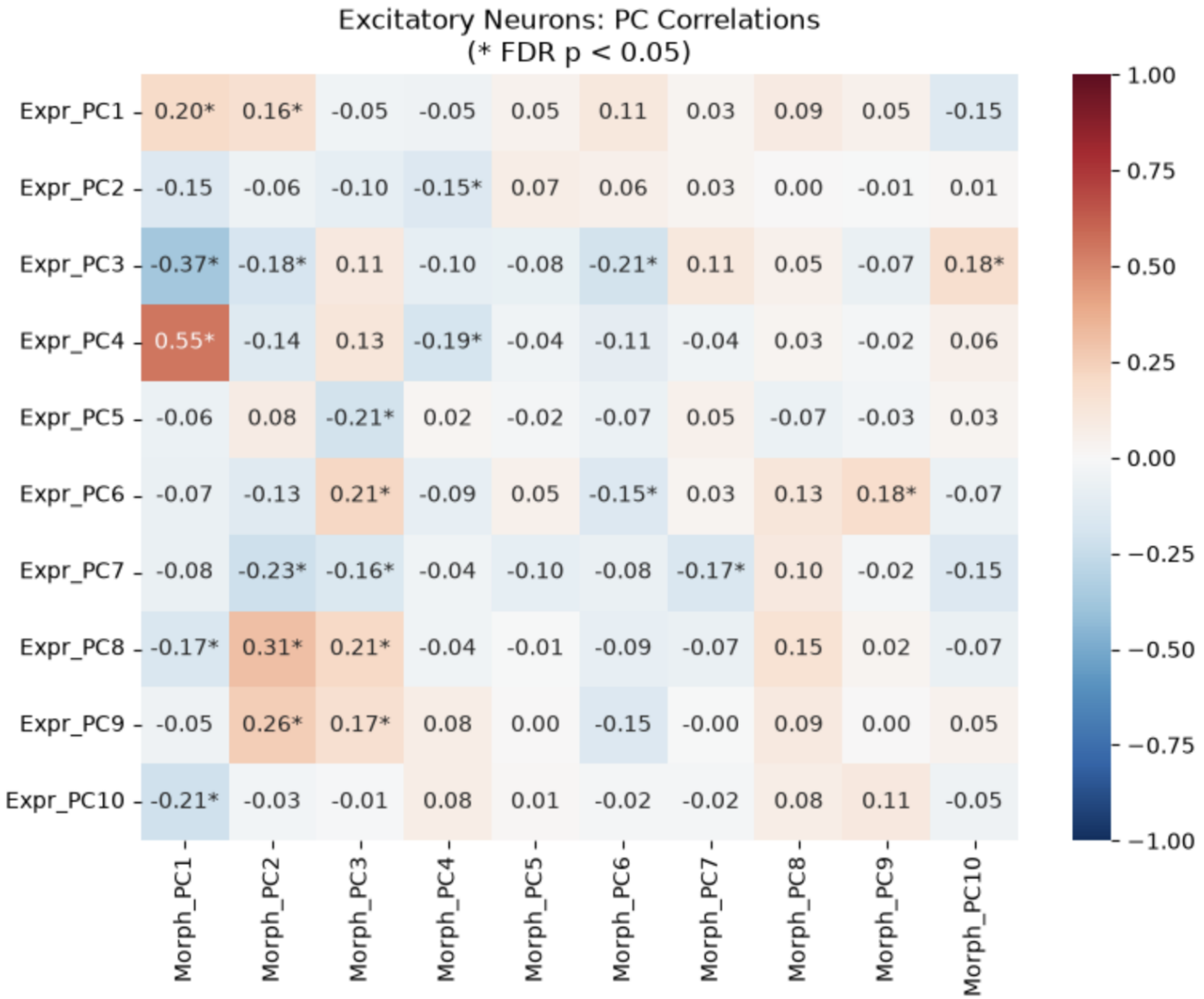
  Morphology indices shuffled
- Multiple testing correction

  Benjamini-Hochberg FDR correction
- Heatmap across PC1-10 for both analyses

Highest correlation

Expression PC4 and Morphology PC1

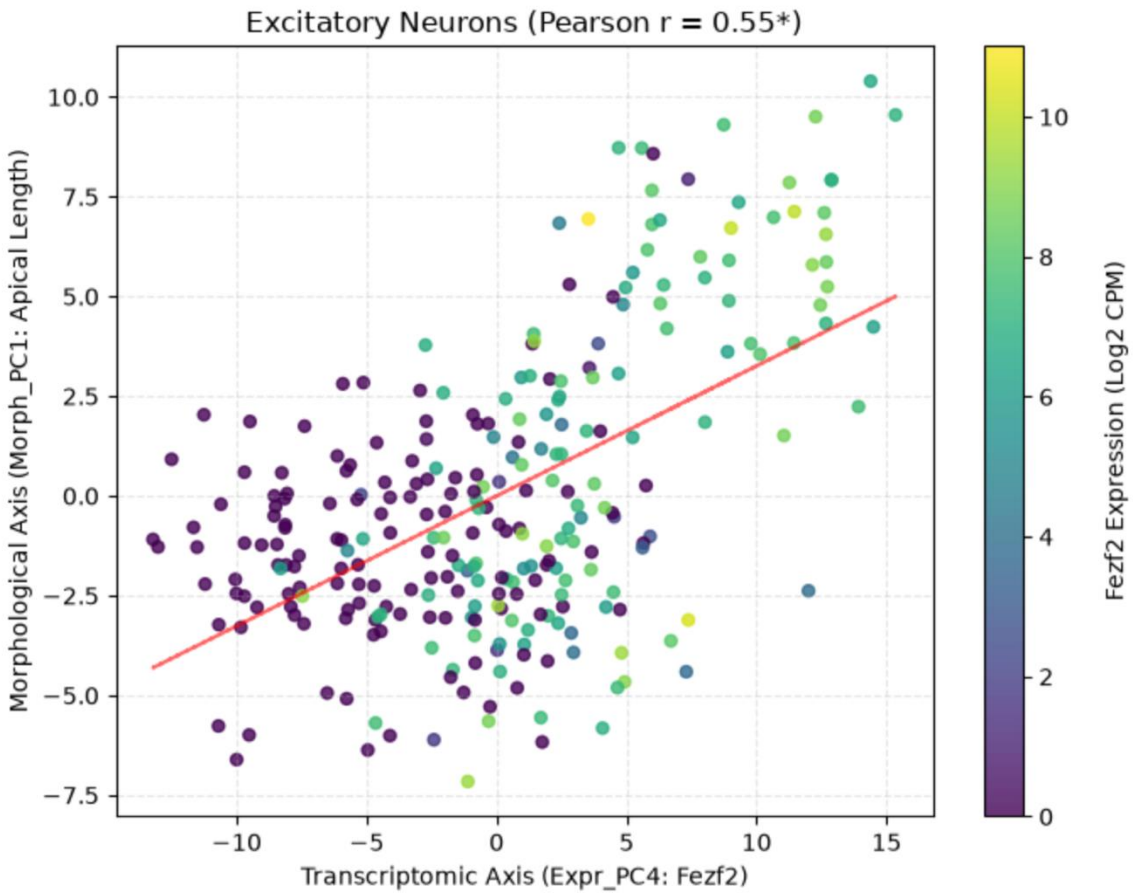
r=0.55, adjusted p<0.05 following FDR correction



Pairwise PC Correlation

```
Top genes driving PC4
Positive drivers (associated with high Morph_PC1):
Fezf2      0.107321
Coro6      0.094253
Tox        0.090083
Tox2       0.081522
Gm49948    0.080009
Parm1      0.079595
Flt3       0.078279
Lgr5       0.076358
Gm23153    0.074453
Nkain3     0.074379
```

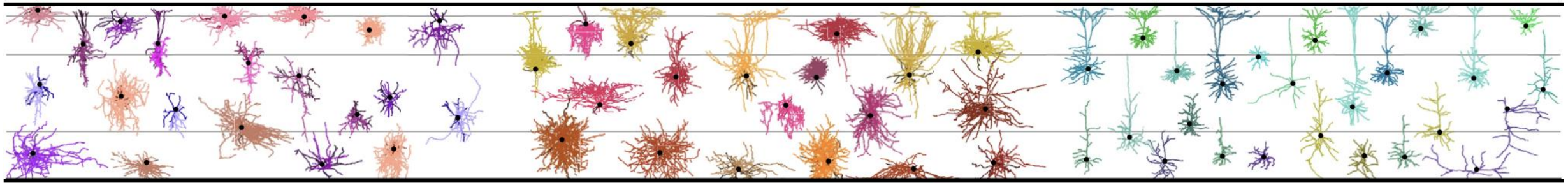
```
Top morph features driving PC1:
Positive structural features:
"apical" total length                  0.248434
dendrite max path distance to soma     0.246348
dendrite max Euclidean dist            0.242465
"apical" robust height                 0.241888
"apical" height                        0.240342
```



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Conclusions



FINDINGS

- Biologically relevant gene expression changes correlated with associated morphological features.
- Statistical validation via Permutation Testing and FDR ensured observed correlations were biologically significant

LIMITATIONS

- Lack of accessibility: SWC raw files unavailable, berenslab/mini-atlas pipeline dependent on raw data
- Causality vs Correlation: genes and morphologies may be correlated but not *caused by* one another.

IMPROVEMENTS

- Compare against Canonical Correlation Analysis (CCA)
- GSEA of the returned top-loading genes (e.g., gseapy)
- Perform target gene-level regressions on top candidate genes.

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References

Guo, C *et al.* Fezf2 expression identifies a multipotent progenitor for neocortical projection neurons, astrocytes, and oligodendrocytes. *Neuron* **80(5),** 1167-74 (2013).

Kobak, D. & Berens, P. The art of using t-SNE for single-cell transcriptomics. *Nat. Commun*. **5416**, 10 (2019).

Peng, R *et al.*, A generative framework for predicting cellular morphological and transcriptomic perturbation responses. *Cell Reports Methods* ***6,*** *7* (2026).

Scala, F., Kobak, D., Bernabucci, M. *et al.* Phenotypic variation of transcriptomic cell types in mouse motor cortex. *Nature* **598**, 144–150 (2021).

https://github.com/berenslab/mini-atlas

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